ARLON ELECTRONIC MATERIALS

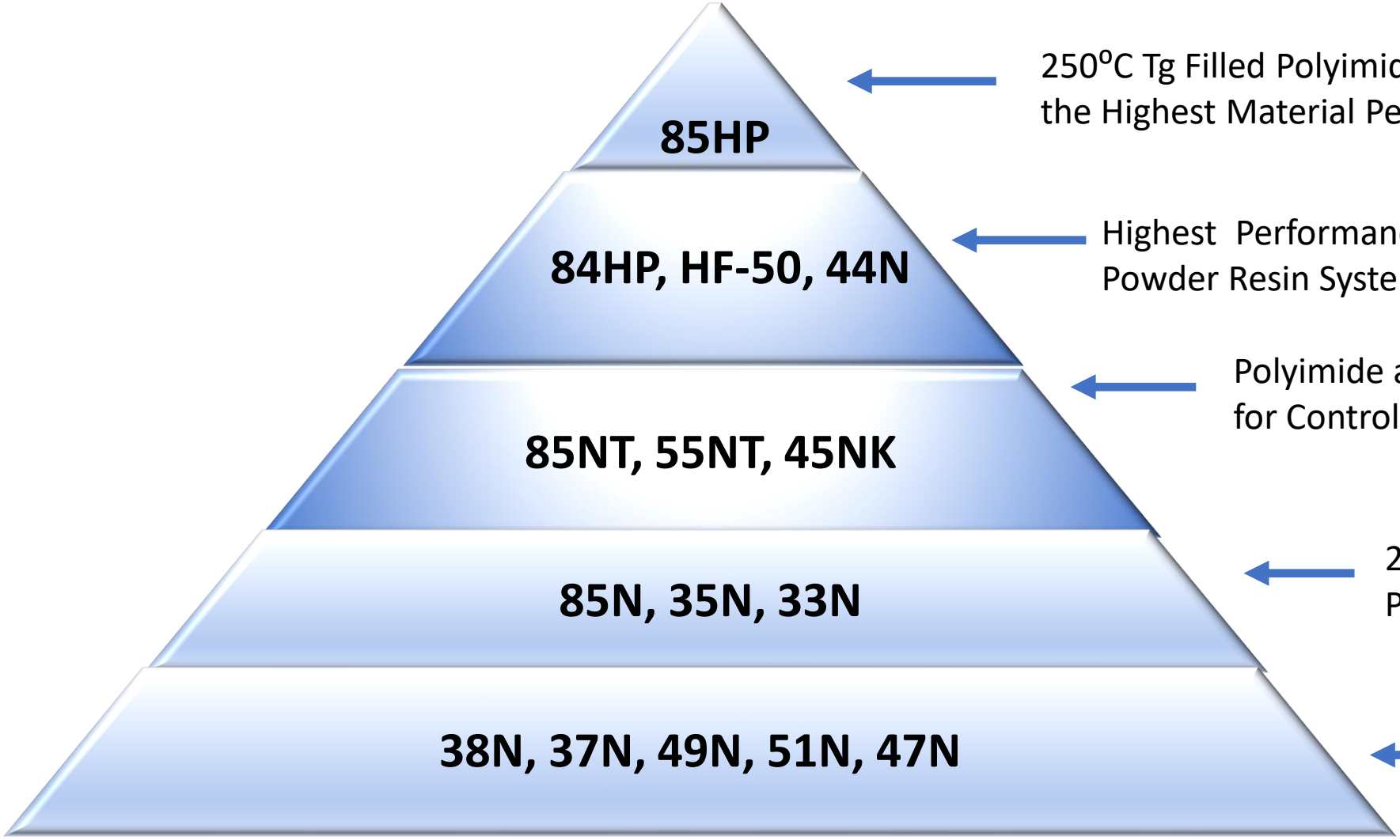
An EMC Company

85HP/84HP Product Introduction

June 2021

Company Confidential

Best in Class Thermoset Products



250°C Tg Filled Polyimide Resin System with the Highest Material Performance Available

Highest Performance Filled Prepreg and Powder Resin Systems

Polyimide and Epoxy Resin systems for Controlled X/Y CTE <9 ppm/°C

250°C Tg Non- Filled Polyimide Systems

High Performance Polyimide and Epoxy Low Flow Prepregs

85HP High Performance Filled Polyimide Resin System

85HP a **pure polyimide** resin which incorporates a **proprietary filler system,** benefits are:

- Lower Z axis expansion (<1% between 50-260º)
- Improved hole wall drilling, reducing tear out and copper wicking in glass bundles
- Improved dimensional stability with balanced weave glass and proprietary filler
- 2.5X higher thermal conductivity than other polyimide systems for dissipating heat
- Reduces risk of resin cracking in neat resin areas during thermal excursions and mechanical drilling
- Allows for > 5X sequential lamination cycles and higher layer count MLB's
- Bromine and Halogen free resin system

Typical Applications for 85HP/84HP Polyimide Printed Wiring Boards:

- High process or assembly temperatures (lead-free soldering)
- Designs with high layer counts and MLB complexity using multiple lamination cycles
- PWB's employing copper weights over 2 ounces opposing
- Equipment exposed to extreme temperatures
  - Defense systems
  - Aircraft engine instrumentation
  - Semiconductor testing (burn-in boards)
  - Petroleum exploration (down hole drill heads)
  - Under-hood automotive
  - Industrial sensor systems
  - Deep Space and satellites

IPC Slash Sheets and Test Specifications

IPC-4101E-WAM1 Slash Sheets

- Slash sheets define and identify the constituency and performance requirements of laminates and prepregs.
- IPC-4101 slash sheets are important to allow the OEM or designer to compare material properties to the end application.
- 85HP and 84HP for example are slashed to 40, 41 and 43, each slash sheet has its own material properties.
- In the case of 85HP slashed to 43, the designer requires a 250°C Tg and a low Z-axis CTE of 1% or less between 50°C-260°C.

IPC-4101E-WAM1 Slashed Sheets

- Material properties are used to determine how a material is slashed by IPC (Note: No Secondary Resin System with 40, 41 and 43)

SPECIFICATION SHEET #: IPC-4101/40 **REINFORCEMENT:** *1:* Woven E-glass **RESIN SYSTEM:** ***Primary:*** Polyimide ***Secondary 1:*** NONE **FLAME RETARDANT MECHANISM:** AABUS **FILLERS:** With or without inorganic fillers **ID REFERENCE:** ***UL/ANSI:*** GPY ***ANSI:*** GPY/40 **GLASS TRANSITION (T_g):** 200 °C minimum	*2:* NONE ***Secondary 2:*** NONE ***Minimum UL94 Requirement:*** HB ***MIL-S-13949:*** /10 - GI, GIN, GIJ, GIP, GIL	**Keywords: (For Search Only)** **NOT Grade Requirement** **See Section 7** Polyimide / Woven Glass High Reliability GPY Fillers
SPECIFICATION SHEET #: IPC-4101/41 **REINFORCEMENT:** *1:* Woven E-glass **RESIN SYSTEM:** ***Primary:*** Polyimide ***Secondary 1:*** NONE **FLAME RETARDANT MECHANISM:** AABUS **FILLERS:** With or without inorganic fillers **ID REFERENCE:** ***UL/ANSI:*** GPY ***ANSI:*** GPY/41 **GLASS TRANSITION (T_g):** 250 °C minimum	*2:* NONE ***Secondary 2:*** NONE ***Minimum UL94 Requirement:*** HB ***MIL-S-13949:*** /10 - GIL, GIP	**Keywords: (For Search Only)** **NOT Grade Requirement** **See Section 7** Polyimide / Woven Glass High Reliability GPY Fillers
SPECIFICATION SHEET #: IPC-4101/43 **REINFORCEMENT:** *1:* Woven E-glass **RESIN SYSTEM:** ***Primary:*** Polyimide ***Secondary 1:*** NONE **FLAME RETARDANT MECHANISM:** AABUS **FILLERS:** Contains inorganic fillers **ID REFERENCE:** ***UL/ANSI:*** NONE ***ANSI:*** NONE **GLASS TRANSITION (T_g):** 250 °C minimum	*2:* NONE ***Secondary 2:*** NONE ***Minimum UL94 Requirement:*** HB ***MIL-S-13949:*** /10 - GI, GIN, GIL, GIP	**Keywords: (For Search Only)** **NOT Grade Requirement** **See Section 7** Polyimide / Woven Glass Non-GPY High Reliability Low Moisture Absorption Low Loss Tangent Low Z-axis CTE Greater Time to Delaminate Fillers

IPC-4101E-WAM-1 Slash Sheets

Arlon's 85HP meets the requirements of the three slash sheets, 40, 41 and 43. 85HP is the only polyimide on the market that is slashed to 43 and differentiates this product from all others with 1% or less Z-Axis expansion!

IPC-4101E /43

LAMINATE REQUIREMENTS (continued)					
Laminate Requirement	Specification <0.50 mm [0.0197 in]	Specification ≥0.50 mm [0.0197 in]	Units	Test Method	Ref. Para.
15. Z-Axis CTE A. Alpha 1, maximum B. Alpha 2, maximum C. 50 to 260 °C, maximum (Total Expansion)	 – – –	 40 175 1.0	 ppm/°C ppm/°C %	2.4.24	3.10.1.11
16. Time to Delaminate (Copper removed) A. T260, minimum B. T288, minimum C. T300, minimum	 – – –	 60 15 5	Minutes	2.4.24.1 & as modified in 3.10.1.12	3.10.1.12
17. Other	–	–			

Comparison of Properties of 85N/85HP Polyimide

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85HP

Typical Properties:

Property	Units	Value	Test Method
Electrical Properties			
Dielectric Constant @ 1 MHz	@ 54% RC	4.2	IPC TM-650 2.5.5.9
Dissipation Factor @ 1 MHz	@ 54% RC	0.009	IPC TM-650 2.5.5.9
Volume Resistivity			
C96/35/90	MΩ-cm	8.0×10^7	IPC TM-650 2.5.17.1A
E24/125	MΩ-cm	5.9×10^7	IPC TM-650 2.5.17.1A
Surface Resistivity			
C96/35/90	MΩ	1.6×10^9	IPC TM-650 2.5.17.1A
E24/125	MΩ	6.9×10^7	IPC TM-650 2.5.17.1A
Electrical Strength	Volts/mil (kV/mm)	1208 (48)	IPC TM-650 2.5.6.2A
Arc Resistance	sec	182	IPC TM-650 2.5.1B
Thermal Properties			
Glass Transition Temperature (Tg)			
TMA	°C	≥250	IPC TM-650 2.4.24C
Decomposition Temperature			
Initial	°C	[illegible]	IPC TM-650 2.4.24.6
5% weight loss	°C	430	IPC TM-650 2.4.24.6
T260	min	>60	IPC TM-650 2.4.24.1
T288	min	>60	IPC TM-650 2.4.24.1
T300	min	>60	IPC TM-650 2.4.24.1
CTE (X,Y)	ppm/°C	17, 17	IPC TM-650 2.4.41
CTE (Z)			
< Tg	ppm/°C	38.8	IPC TM-650 2.4.24C
> Tg	ppm/°C	154	IPC TM-650 2.4.24C
z-axis Expansion (50-260°C)	%	≤1	IPC TM-650 2.4.24C
Mechanical Properties			
Peel Strength to Copper (1 oz/35 micron)			
After Thermal Stress	lb./in (N/mm)	6.3 (1.11)	IPC TM-650 2.4.8C
At Elevated Temperatures	lb./in (N/mm)	5.2 (.92)	IPC TM-650 2.4.8.3A
After Process Solutions	lb./in (N/mm)	5.3 (.93)	IPC TM-650 2.4.8C
Young's Modulus CD/MD	Mpsi (GPa)	3.0	ASTM E111
Tensile Strength CD/MD	kpsi (MPa)	35 (241)	ASTM D3039
Poisson's Ratio	-	0.15	ATM E13204
Physical Properties			
Water Absorption (0.059")	%	0.19	IPC TM-650 2.6.2.1A
Density	g/cm3	1.7	ASTM D792 Method A
Thermal Conductivity	W/mK	0.5	ASTM E1461
Flammability	class	HB	UL-94

Results listed above are typical properties, provided without warranty, expressed or implied, and without liability. Properties may vary, depending on design and application. Arlon reserves the right to change or update these values.

85N

Typical Properties:

Property	Units	Value	Test Method
Electrical Properties			
Dielectric Constant @ 1 MHz	-	4.2	IPC TM-650 2.5.5.3
@ 1 GHz	-	4.0	
Dissipation Factor @ 1 MHz		0.01	IPC TM-650 2.5.5.3
@ 1 GHz		N/A	
Volume Resistivity			
C96/35/90	MΩ-cm	1.5×10^8	IPC TM-650 2.5.17.1
E24/125	MΩ-cm	3.0×10^{10}	IPC TM-650 2.5.17.1
Surface Resistivity			
C96/35/90	MΩ	1.6×10^9	IPC TM-650 2.5.17.1
E24/125	MΩ	1.6×10^9	IPC TM-650 2.5.17.1
Electrical Strength	Volts/mil (kV/mm)	1450 (57.1)	IPC TM-650 2.5.6.2
Dielectric Breakdown	kV	>40	IPC TM-650 2.5.6
Arc Resistance	sec	143	IPC TM-650 2.5.1
Thermal Properties			
Glass Transition Temperature (Tg)			
TMA	°C	=>250	IPC TM-650 2.4.24
DSC	°C		IPC TM-650 2.4.25
Decomposition Temperature			
Initial	°C	387	IPC TM-650 2.4.24.6
5% weight loss	°C	407	IPC TM-650 2.4.24.6
T260	min	>60	IPC TM-650 2.4.24.1
T288	min	>60	IPC TM-650 2.4.24.1
T300	min	>60	IPC TM-650 2.4.24.1
CTE (X,Y)	ppm/°C	16	IPC TM-650 2.4.41
CTE (Z)			
< Tg	ppm/°C	55	IPC TM-650 2.4.24
> Tg	ppm/°C	149	IPC TM-650 2.4.24
z-axis Expansion (50-260°C)	%	1.2	IPC TM-650 2.4.24
Mechanical Properties			
Peel Strength to Copper (1 oz/35 micron)			
After Thermal Stress	lb./in (N/mm)	6.4 (1.1)	IPC TM-650 2.4.8
At Elevated Temperatures	lb./in (N/mm)	6.4 (1.1)	IPC TM-650 2.4.8.2
After Process Solutions	lb./in (N/mm)	6.4 (1.1)	IPC TM-650 2.4.8
Young's Modulus CD/MD	Mpsi (GPa)	3.6/4.1 (24.8/28.2)	ASTM E111
Tensile Strength CD/MD	kpsi (MPa)	48/64 (330/440)	ASTM D3039
Poisson's Ratio	-	0.18	ASTM E13204
Physical Properties			
Water Absorption (0.062")	%	0.27	IPC TM-650 2.6.2.1
Density	g/cm³	1.6	ASTM D792 Method A
Thermal Conductivity	W/mK	0.2	ASTM E1461
Flammability	class	HB	UL-94

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VALIDATION ANNOUNCEMENT

Arlon EMD is pleased to announce that its 85HP proprietary filled polyimide resin system for cores and prepregs has received official validation from IPC that this resin system meets all requirements of IPC-4101E-WAMI/43.

85HP is now validated to IPC4101/40, 41 and 43!

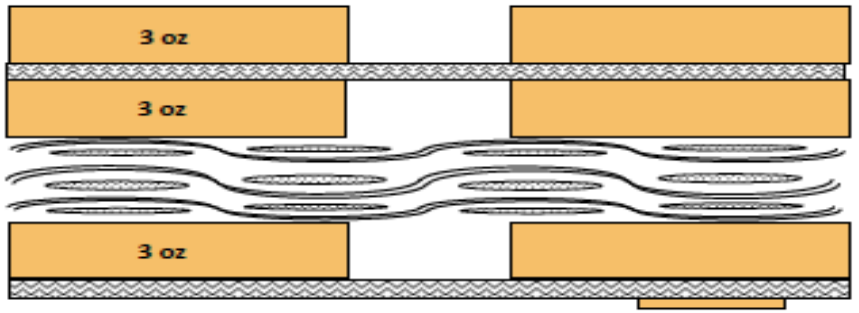
What does this mean for you our valued customers?

> Confidence that your most demanding designs that are; high layer count, thick copper inner layer, filled clearance hole needs and/or requires multiple lamination cycles can be reliably manufactured.

A few standout properties of the proprietary 85HP resin system:

- <1% z-axis expansion from 50-260°
- 2.5X higher thermal conductivity than other polyimide systems for dissipating heat
- Longer time to delamination - T300 >60 minutes
- Allows for > 5X sequential lamination cycles and higher layer count MLB's
- Improved mechanical and laser drilling

85HP Construction



Contact Arlon today to discuss your Challenges!

85N/85HP Product Overview

85HP

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High Performance Polyimide

85HP's unique blend of pure polyimide resin and micro-fine proprietary fillers results in superior performance for demanding applications. Compared to conventional polyimide systems, 85HP has a lower Z-axis expansion and twice the thermal conductivity. 85HP reduces resin cracking and wicking in designs with high density plated through holes and vias. 85HP prepreg has resin flow characteristics and pressed thickness matching standard polyimides.

Features:

- **Meets IPC4101/40,41 and /43 description and specification**
  - Pure polyimide, no secondary resin
  - No epoxy added, blended or reacted

- Best-in-Class thermal properties
  - Tg=> 250°C
  - Decomposition temperature 430°C
  - Tc = 0.5 W/mK, 2x thermal conductivity
  - T300>60 min.
  - Reduced Z-axis expansion
  - Minimizes cracking and wicking
  - Balanced glass weave
  - RoHS/WEEE compliant

85HP Overview

- The use of heavy copper and higher layer counts has spiked significantly over the last few years for high power applications.

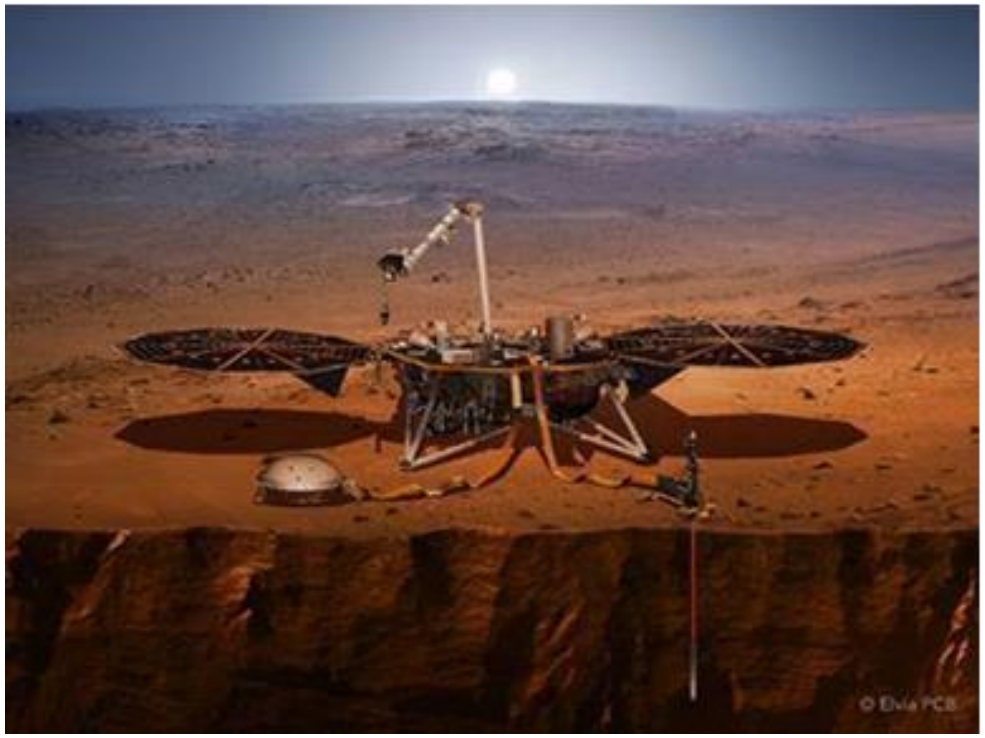
- Processing these boards pose some challenges such as drilling, plating, thermal expansion, resin voids, resin cracking, board warping.

Arlon's solution was the creation of 85HP and 84HP

- The benefits of 85HP are improved via and through hole drilling.
- Reduction in copper wicking.
- Consistent etch back.
- Reduced the propensity of resin cracking.
- Reduced resin shrinking.
- Low Z-axis expansion of less than 1%.
- Recently released 84HP uses the same 85N resin with a micro-fine proprietary filler coated on 50" 106 glass and matches the grain direction of the 85HP.

85HP/84HP OEM's and Applications

- Lockheed Martin Corporation- Deep Space and Aerospace
- Northrop Grumman- Aerospace
- Raytheon Corporation- Military (naval, aerospace, missiles)
- Schlumberger- Down Hole Drilling
- NASA- Space/Ground-Based Systems
- Southwest Research Institute- Space and Launch Systems
- United Launch Alliance- Space, Launch Systems



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Comparison of 85N and 85HP Pure Polyimide Resin Systems

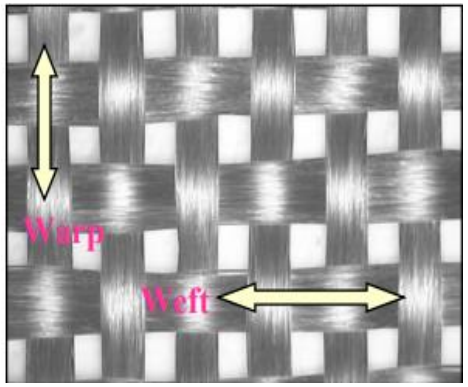
Property	Units	85N	85HP
Dielectric Constant @ 1 MHz	MHz	4.2	4
Dissipation Factor @ 1 MHZ	MHz	0.01	0.008
Glass Transition Temperature (Tg) by TMA	°C	>250	>250
T260	Minutes	>60	>60
T288	Minutes	>60	>60
T300	Minutes	>60	>60
Electrical Strength	Volts/mil	1450	1700
Decomposition Temperature Td (5% weight loss)	°C	407	430
Thermal Conductivity	W/mK	0.20	0.50
Coefficient of Thermal Expansion X-Y Axis	ppm/°C	17	17
Coefficient of Thermal Expansion Z-Axis (<Tg)	ppm/°C	55	40
Coefficient of Thermal Expansion Z-Axis (>Tg)	ppm/°C	150	150
Z-Axis Expansion (50-260°C)	Percent	1.2	<1

85HP Incorporates Balanced Weave Glass

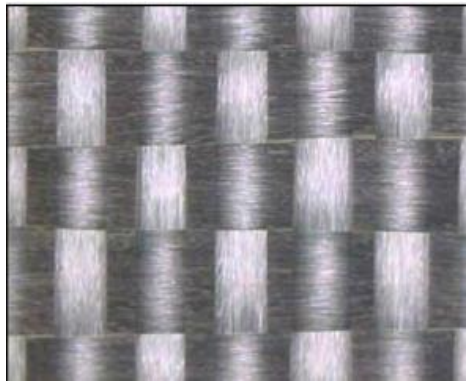
Arlon's 85HP utilizes **balanced** weave fabrics in all core constructions:

- 1067 balanced weave replaces 106 standard glass
- 1086 balanced weave replaces 1080 standard glass
- 2116 balanced weave replaces 2116 standard glass
- 3313 balanced weave replaces 2313 standard glass
- 1506 balanced weave replaces 7628 standard glass

Standard weave



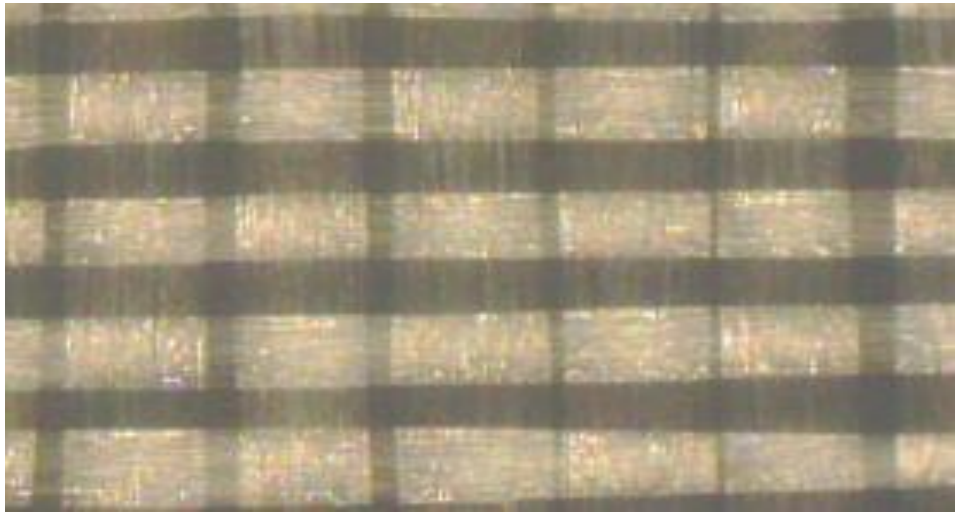
Balanced Weave



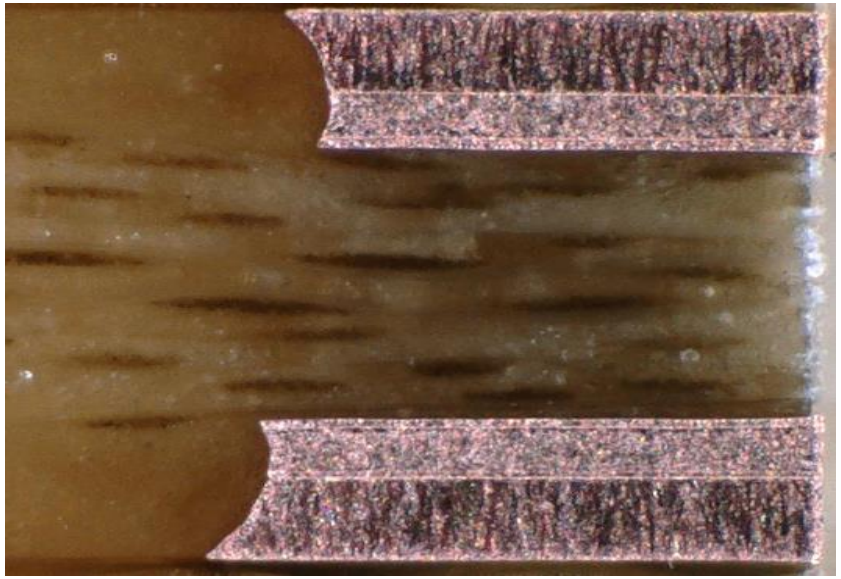
- 85HP is the only polyimide resin system on the market that incorporates balanced weave as a standard product offering to improve electrical and mechanical uniformity!

Benefits of balanced glass styles:

- ✓Reduced open space between the glass yarns
- ✓Uniform layer of glass fibers across the plane
- ✓Improved X, Y, Z- axis CTE Expansion



1067



- ✓Improved drilled through hole and via hole quality
- ✓Reduced copper wicking
- ✓Electrical uniformity is improved
- ✓Flow control and flatness is improved
- ✓More consistent dimensional stability

85HP Constructions using Various Balanced Weave Glass Styles

85HP		1067	1086				3313		2116		1506
Core	Construction Code										
0.0030	A				1						
0.0033	A					1					
0.0033	C	2									
0.0040	A								1		
0.0040	B	1	1								
0.0040	C										
0.0045	A	1			1						
0.0050	A	1					1				
0.0050	B		2								
0.0060	A	1							1		
0.0060	B				2						
0.0072	A		3								
0.0080	A							1	1		
0.0091	A	2			2						
0.0100	A				2				1		
0.0120	B										2
0.0140	A								1	2	
0.0200	A		1								3
0.0200	B				2		4				
0.0210	A				1						3
0.0210	B				3						2
0.0260											
0.0280	A								1		4
0.0300	A										5
0.0390	A						1				6
0.0394 (1mm)	A						1				6
0.0506	A	2	4								6
0.0590	A										10
0.0620	A	1									10

84HP Filled Pure Polyimide Prepreg

84HP High Performance Prepreg

84HP

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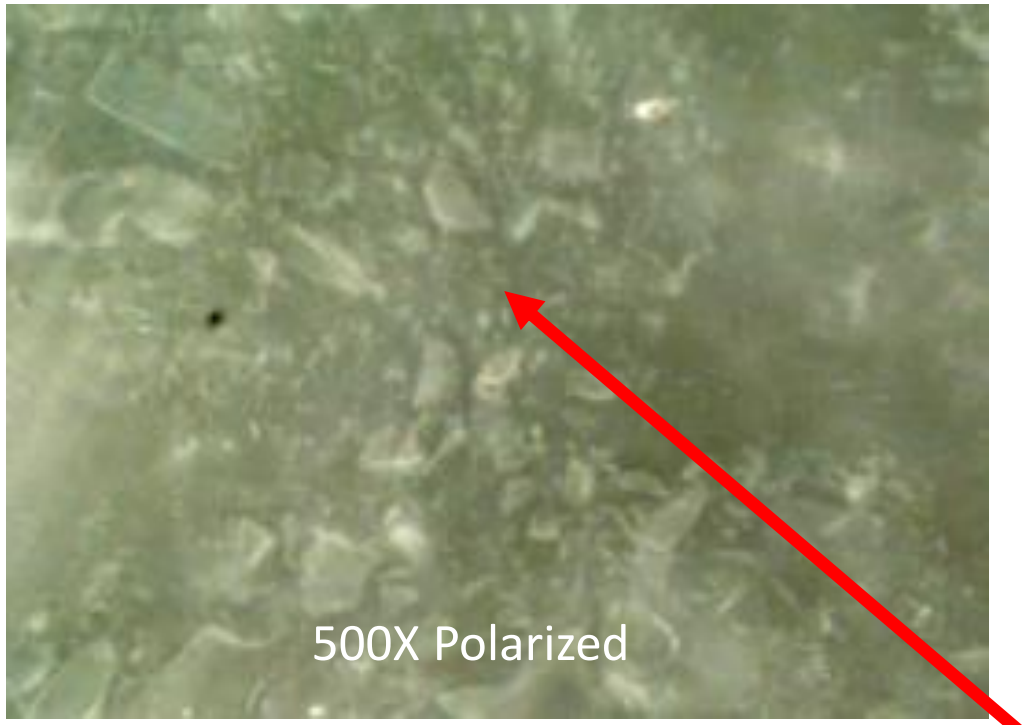
High Performance Polyimide

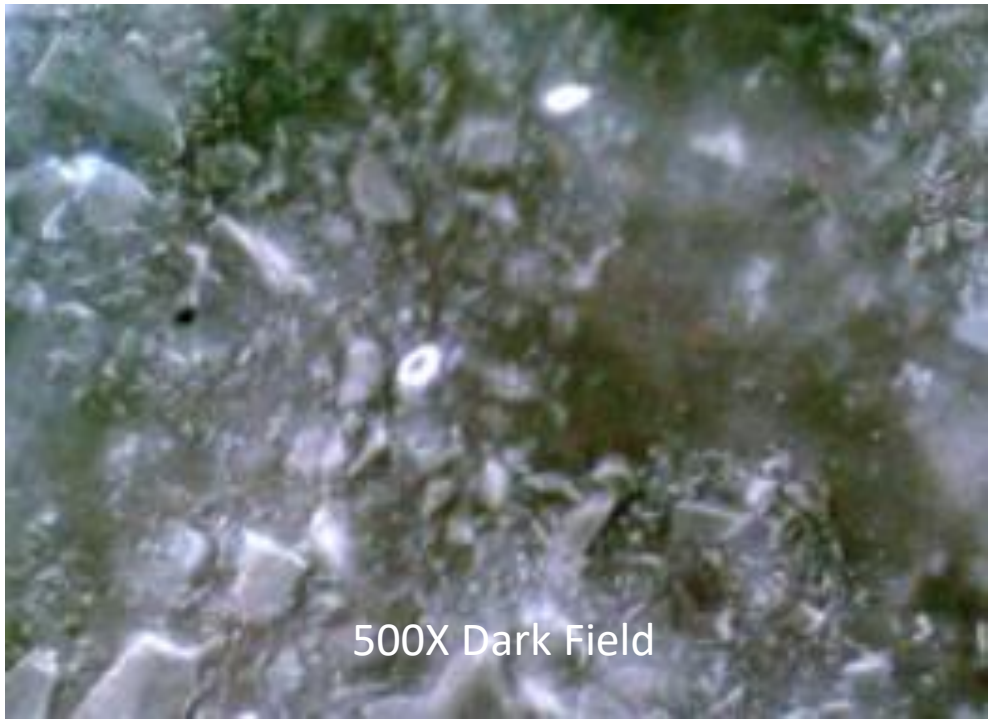
84HP is a high-performance polyimide prepreg with micro-fine proprietary fillers designed for use in filling etched polyimide multilayers that contain thick copper layers and clearance holes in metal cores. 84HP's revolutionary filler system serves to reduce shrinkage and inhibit crack formation during through-hole drill in filled clearance areas. 84HP is matched to copper's CTE at 17 ppm X-Y for controlled expansion during large temperature variations reducing the potential of resin damage due to CTE mismatch.

Features:

- Meets IPC4101/40 and /41 description and specification
  - Pure polyimide, no secondary resin
  - No epoxy added, blended or reacted
- Best-in-Class thermal properties
  - Tg=> 250°C
  - Decomposition temperature 430°C
  - Tc = 0.5 W/mK, 2x thermal conductivity
  - T300>60 min.
- Reduced Z-axis expansion
- Minimizes cracking and wicking
- RoHS/WEEE compliant

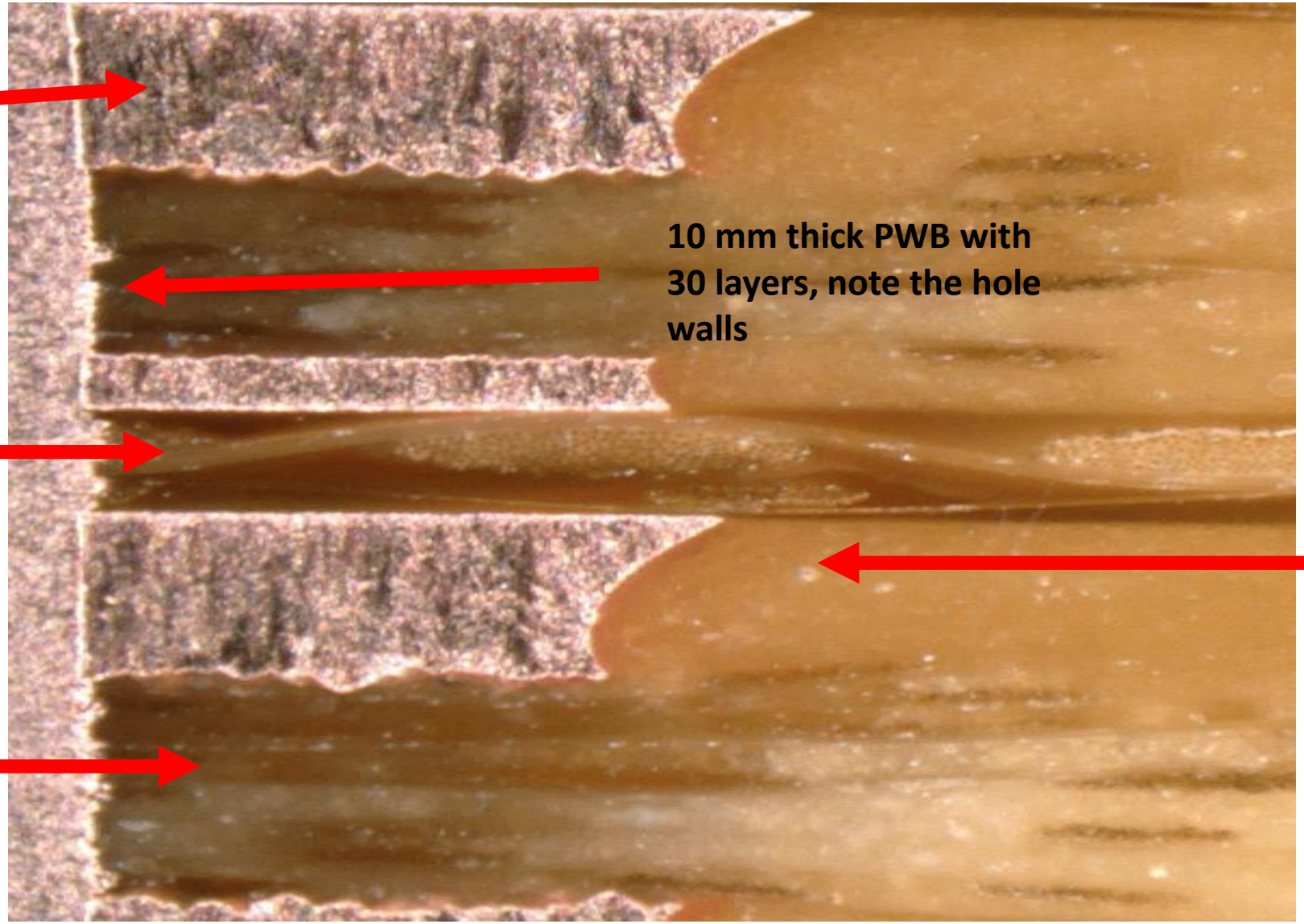
Cross Section of Fillers used in 85HP/84HP



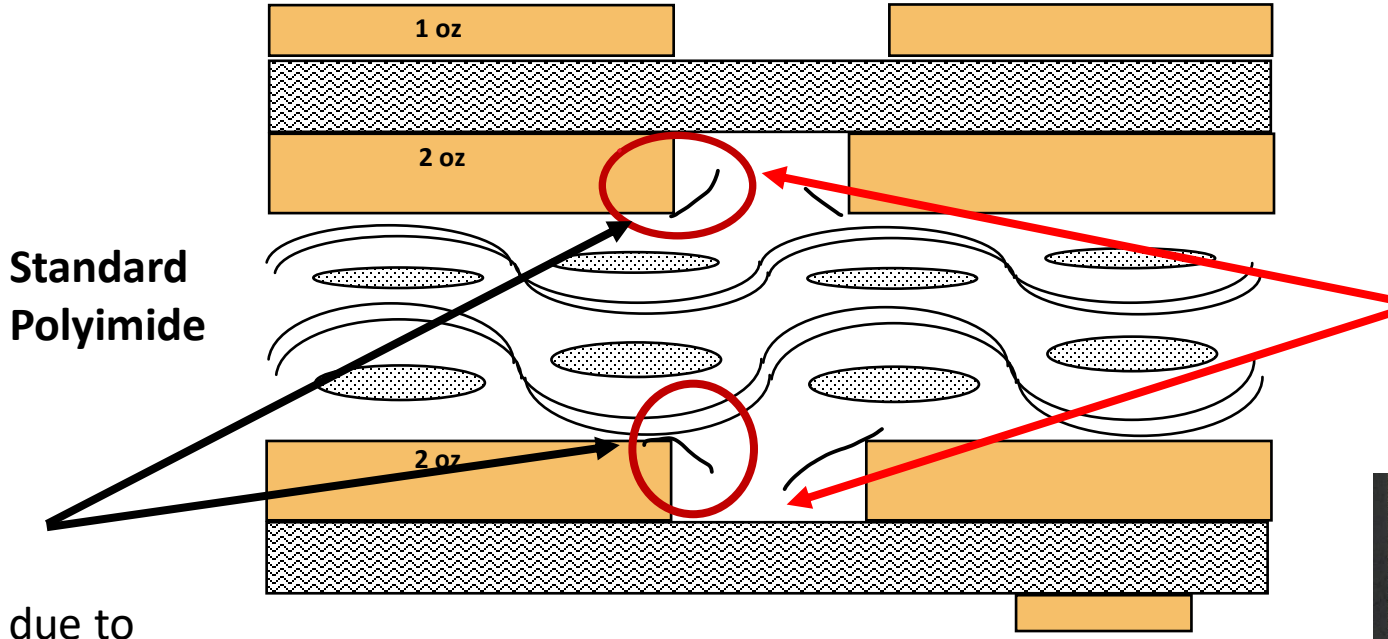


Particle size= 6-20 microns

85HP & 84HP Filling 6-ounce copper layers



Typical Defects in Resin Rich Areas of a PWB

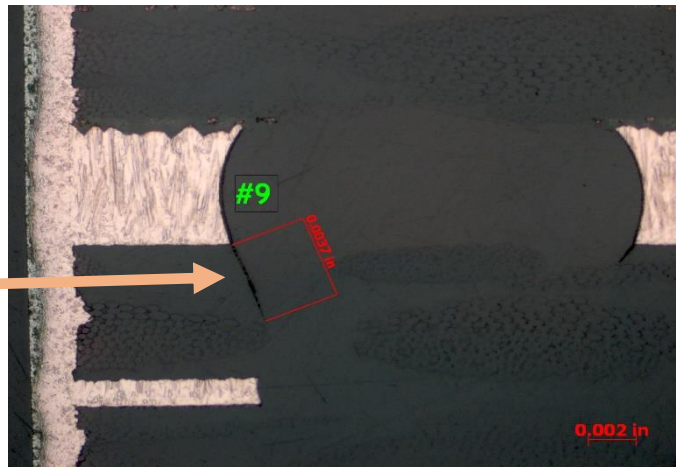


Standard Polyimide

Neat resin areas of a 2 oz over 2 oz PWB (5.2 mils of resin filling between clearance areas)

Tail crack formation due to thermal stress, resin shrinkage

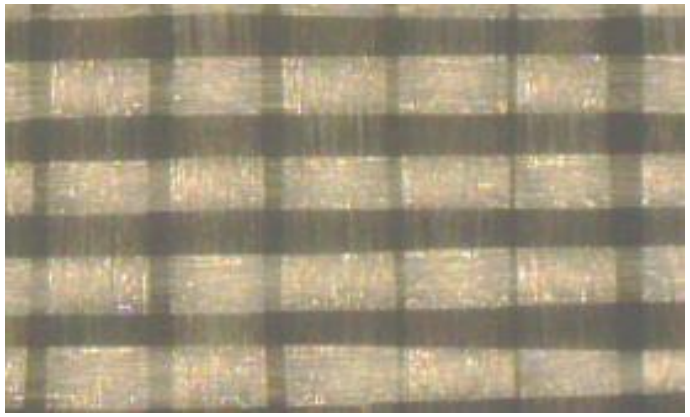
Tail Cracks



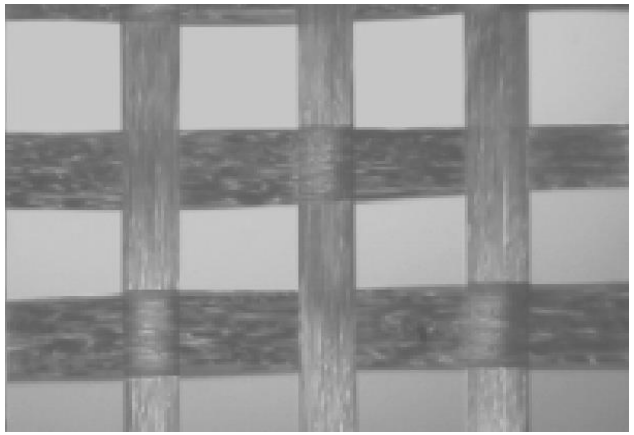
Standard Polyimide Heavy Copper Tail Crack

84HP Design Considerations

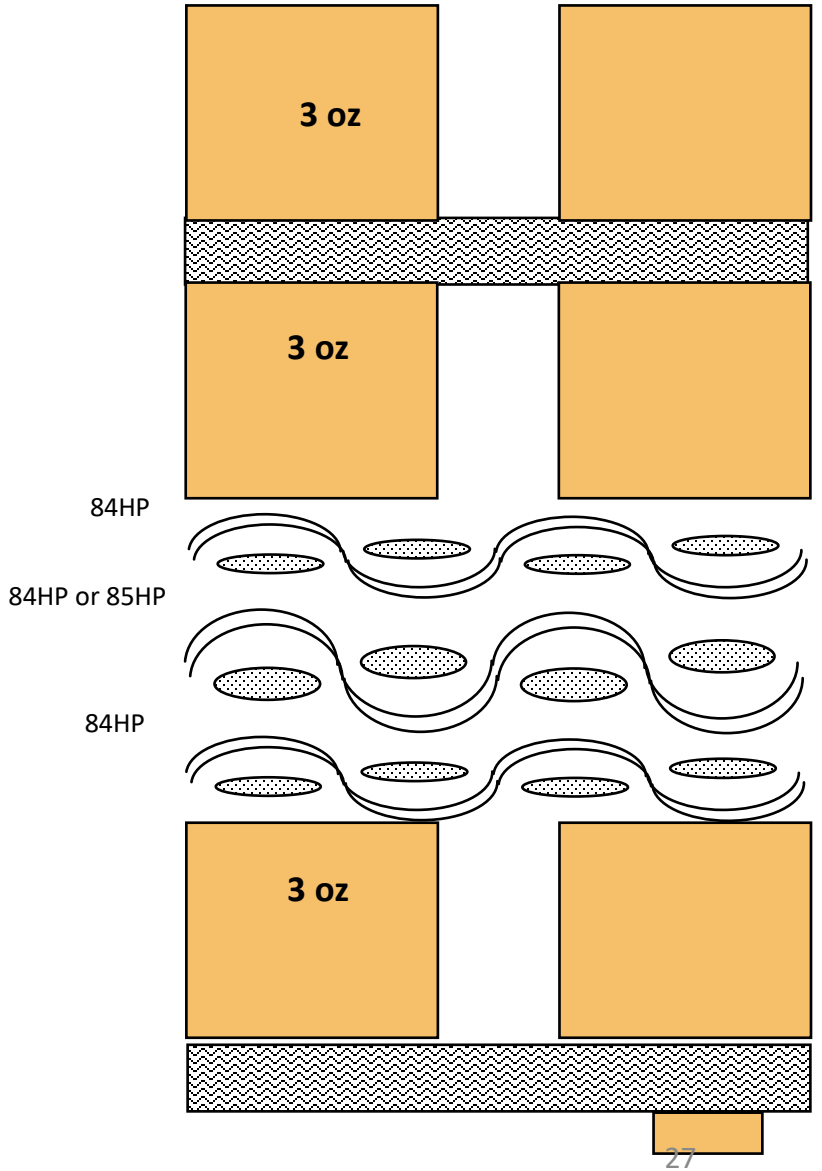
- The **84HP** is necessary on 3 ounce and greater copper thicknesses where there is a large volume of pure neat resin forming in the clearance areas.
- 84HP uses 106 glass to allow the resin to flow into copper circuit patterns.
- Balanced weave glass styles, are employed in the 85HP laminate and prepreg.



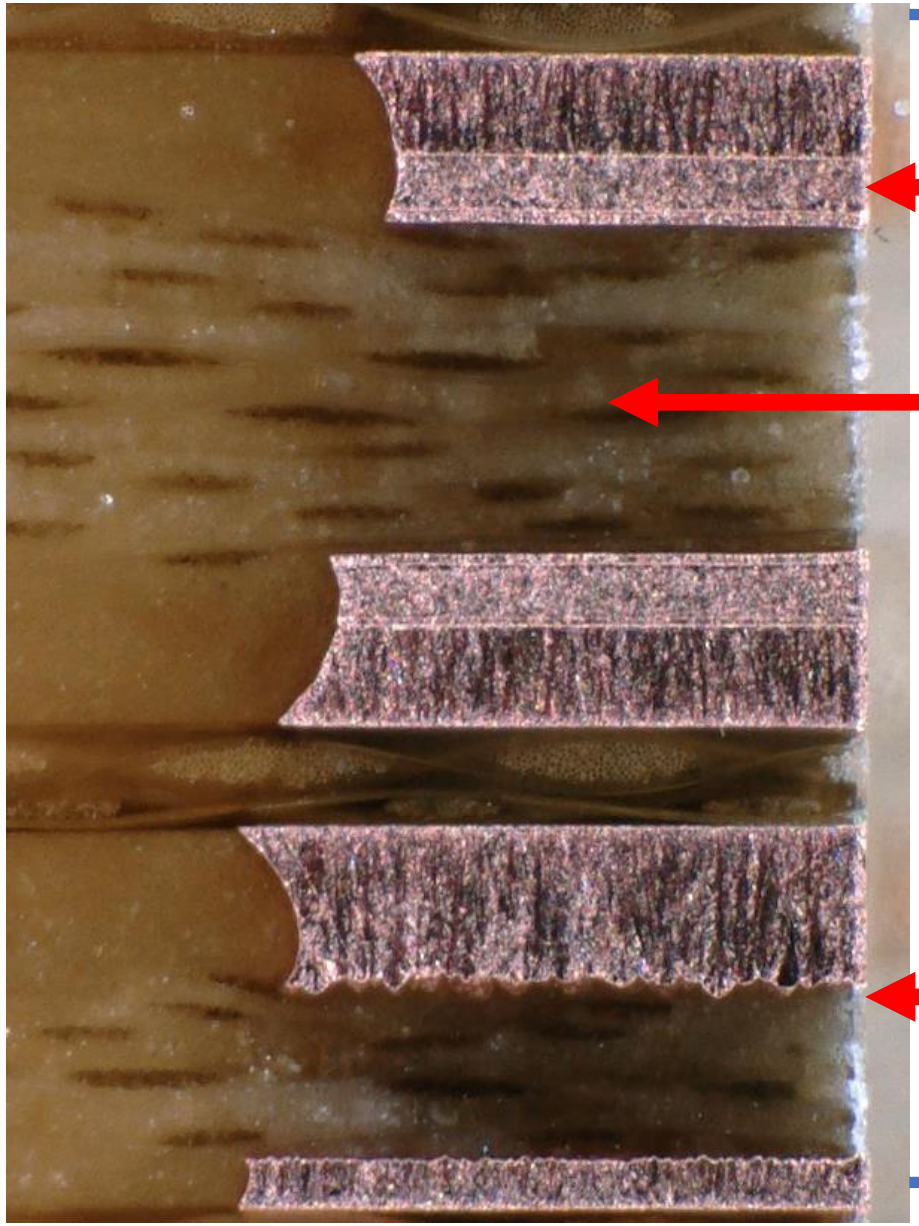
1067



106



85HP/84HP Filling Examples

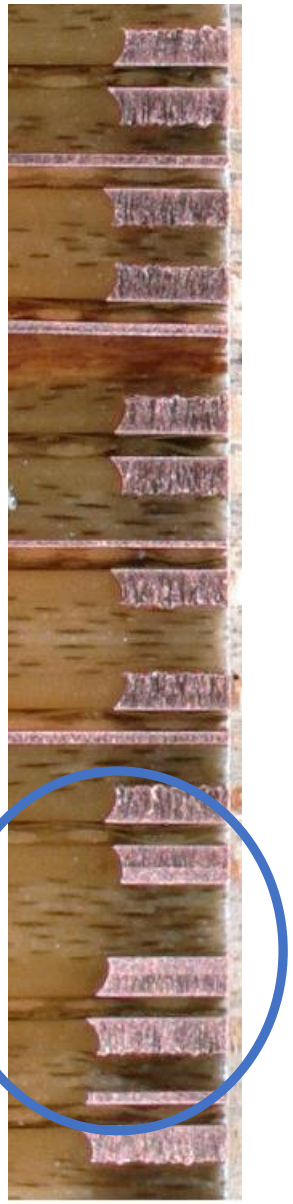


4 Oz copper plating over 6 oz foil

Sequentially Laminated using 84HP and 85HP PP

Peck Drilled holes

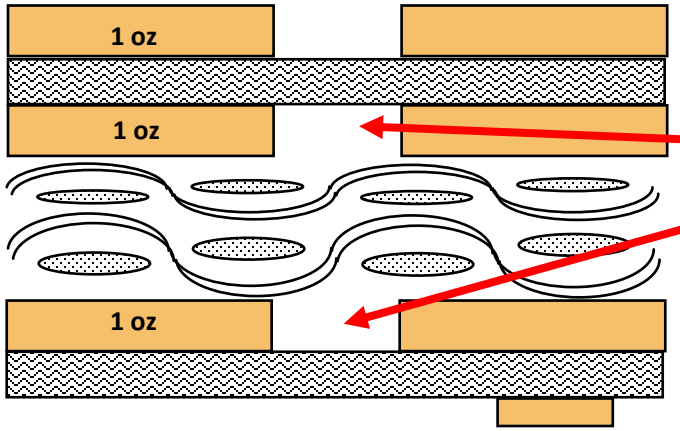
Expanded View



85HP/84HP Design Considerations

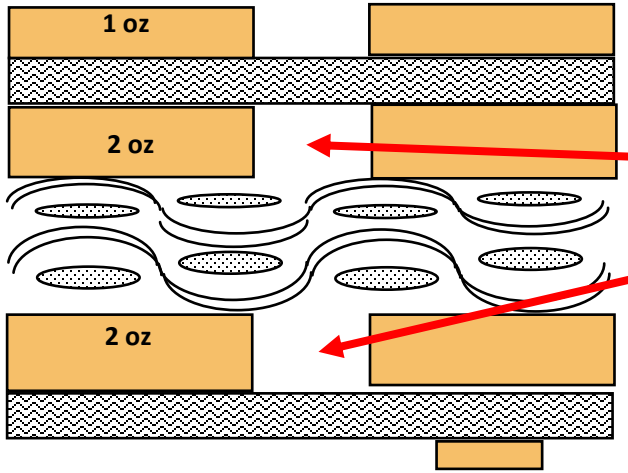
Examples of Pure Resin Rich Areas in Clearance Areas

Standard Polyimide



2.6 mils of neat resin fill with one ounce copper is used

Standard Polyimide

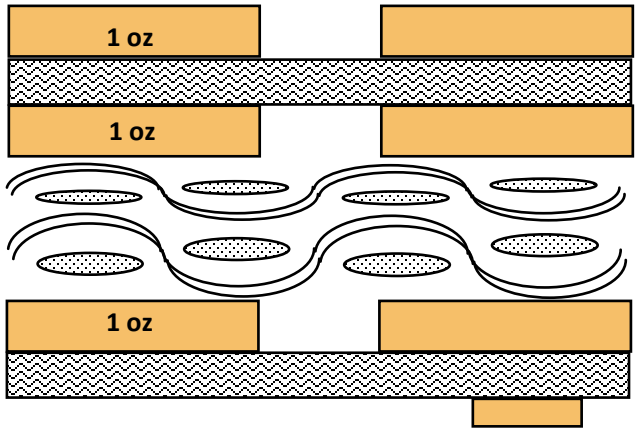


5.2 mils of fill between 2 oz Cu

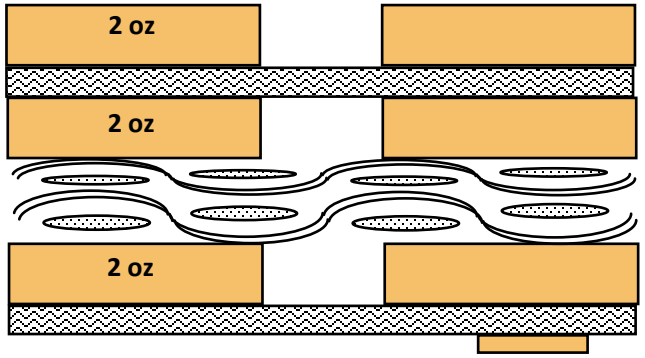
Designing with 85N/85HP/84HP

One- and two-ounce copper constructions typically use standard 85N polyimide without issues, however, high resin rich areas and/or larger drilled hole sizes where there is a concern for resin cracking, customers have chosen 85HP and 84HP prepreg to eliminate these unwanted defects.

Std Poly



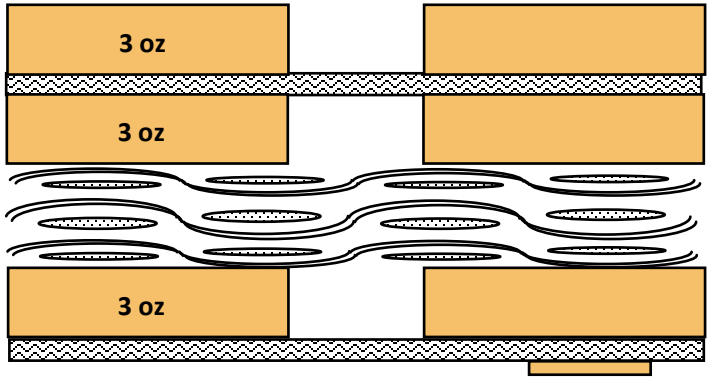
85HP or 84HP



- Copper weights of 3 ounces and higher have been problematic due to CTE and thermal stress related issues causing resin cracking in neat resin areas.

- 85HP and 84HP have proven to reduce or eliminate these issues with the introduction of a proprietary fillers.

84HP
84HP or 85HP
84HP



85N/85HP Processing Guidelines

85HP/84HP PROCESSING VERSUS STANDARD POLYIMIDE

- Scaling: The scale factors do not change unless grain direction or construction changes
  - If you go from 18X24G to 24X18, reverse comps so X factor is now Y factor. Y factor is now X factor

- Prepreg Laminate Storage, Layer Prep & Processing- Standard polyimide recommendations

- Lamination
  - Pre-vac May Need to be Extended
  - Rate of Rise, Cure Cycle-Same
  - Cool-down Rate needs to be $5^{o}F$ or less for Heavy Copper

- Drilling
  - Don't Let the Filler Loading Fool You
  - Unlike other Filled Materials, Drilling is Enhanced
  - Don't Over Peck Heavy Copper PWB's

- Wet Process-Standard polyimide processing recommendations
- Routing-Standard polyimide feeds and speeds
- Assembly Instructions- Standard polyimide processing

Lamination guidelines from Arlon's 85N data sheet

Recommended Process Conditions:

Process inner-layers through develop, etch, and strip using standard industry practices. Use brown oxide on inner layers. Adjust dwell time in the oxide bath to ensure uniform coating. Bake inner layers in a rack for 60 minutes at

107°C - 121°C (225°F - 250°F) immediately prior to lay-up. Store prepreg at 60-70°F at or below 30% RH. Vacuum desiccate the prepreg for 8 - 12 hours prior to lamination.

Lamination Cycle:

1) Pre-vacuum for 30 - 45 minutes

2) Control the heat rise to 4.5°C - 6.5°C (8°F - 12°F) per minute between 100°C and150°C (210°F and 300°F). Vacuum lamination is preferred. Start point vacuum lamination pressures are shown in the table below:

Panel Size		Pressure	
in.	mm	psi	kg/cm2
12 x 18	305 x 457	230	16
16 x 18	406 x 457	300	21
18 x 24	457 x 610	325	23

3) Set cure temperature at 218°C (425°F). Start cure time when product temperature > 213° C (415°F)

4) Cure time at temperature = 120 minutes

5) Cool down under pressure at ≤ 5°C/min (10°F/min)

Drill at 350 SFM. Undercut bits are recommended for vias 0.0 18" (0.45mm) and smaller

De-smear using alkaline permanganate or plasma with settings appropriate for polyimide;

plasma is preferred for positive etchback

Conventional plating processes are compatible with 85N

Standard profiling parameters may be used; chip breaker style router bits are not

recommended Bake for 1 - 2 hours at 250°F (121°C) prior to solder to reflow of HASL

Lamination guidelines from Arlon's 85HP data sheet

Recommended Process Conditions:

Vacuum desiccate the prepreg for 8 - 12 hours prior to lamination.
Process inner-layers through develop, etch, and strip using standard industry practices. Use brown oxide or alternatives on inner layers. Adjust dwell time in the oxide bath to ensure uniform coating. Bake inner layers in a rack for 60 minutes at 107°C - 121°C (225°F - 250°F) immediately prior to lay-up.

Lamination Cycle: (Steps 1-5)

1) Pre-vacuum for 30 - 45 minutes

2) Control the heat rise to 4.5°C - 6.5°C (8°F - 12°F) per minute between 100°C and150°C (210°F and 300°F). Vacuum lamination is preferred. Start point vacuum lamination pressures are shown in the table below:

Panel Size		Pressure	
in.	mm	psi	kg/cm2
12 x 18	305 x 457	275	19
16 x 18	406 x 457	350	25
18 x 24	457 x 610	400	28

3) Product temperature at start of cure = 218°C (425°F)

4) Cure time at temperature = 2.0 hours

5) Cool down under pressure at ≤ 5°C/min (10°F/min)

Drill at 350-400 SFM. Undercut bits are recommended for vias 0.0 18" (0.45mm) and smaller De-smear using alkaline permanganate or plasma with settings appropriate for polyimide; plasma is preferred for positive etchback

Conventional plating processes are compatible with 85HP

Standard profiling parameters may be used;

Bake for 1 - 2 hours at 250°F (121°C) prior to solder to reflow of HASL

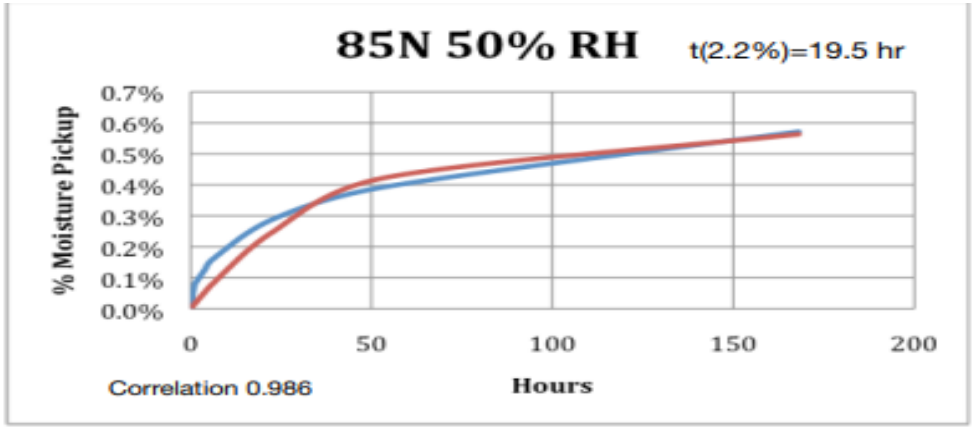
Moisture Absorption of Polyimide MLB's

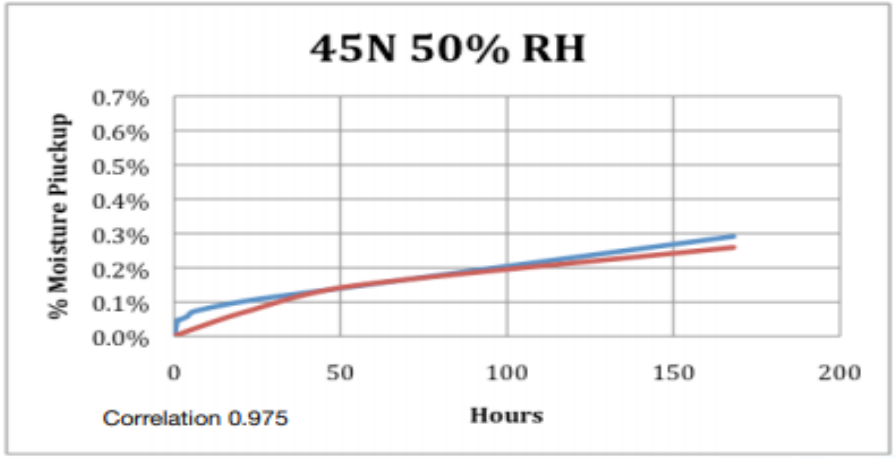
- Polyimide resins tend to absorb moisture faster than other resin systems
- Removing moisture before processing is critical in high humidity conditions
- It has been determined that as little as 0.22% moisture can cause blistering or delamination in PI resins
- Both moisture pickup and subsequent drying are impacted by several variables such as PWB thickness, construction and finished coatings
- A polyimide board may get shipped without issues, but the assembly operation without precautions may run into blistering issues, the chart below illustrates how long boards can sit before they reach 0.22%

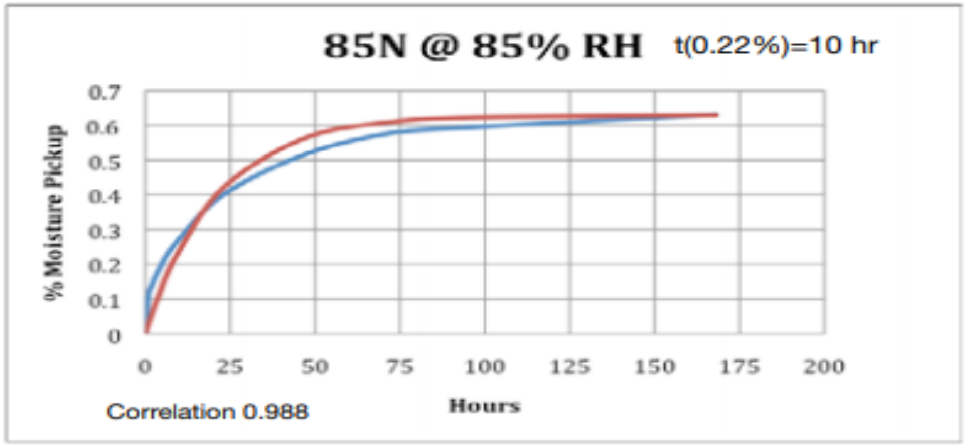
	45N	35N	85N	85HP
85% RH				
Equilibrium	0.28%	0.63%	0.68%	0.52%
Time to 0.22%	55 hr	9 hr	10 hr	12.5 hr
50% RH				
Equilibrium	0.29%	0.56%	0.57%	0.49%
Time to 0.22%	110 hr	20 hr	19.5 hr	33 hr

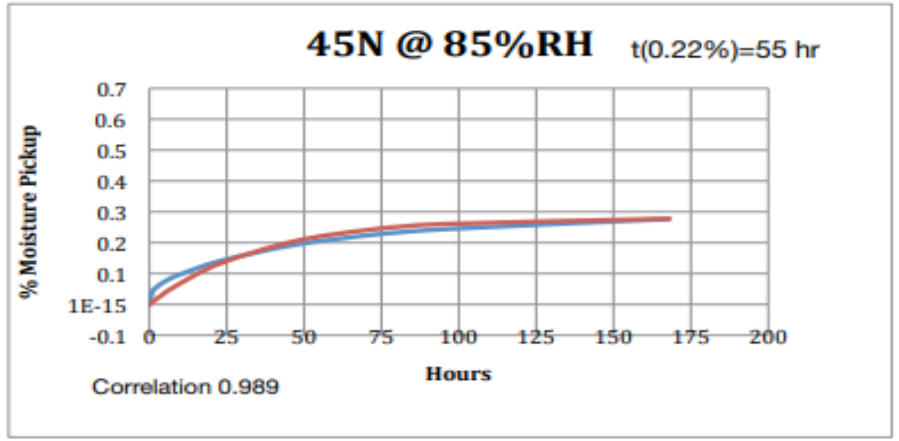
Moisture Absorption of Polyimide MLB's

Moisture Pickup – 85% RH, 25°C
(Blue Line = Original Data, Red Line = Best Fit)









Arlon's Stack-Up Calculator

Technical Sales and Applications Engineers will assist in determining the best and most economical stack up for your challenging design!

Example of Arlon's Stack-up Calculator

ELECTRONIC MATERIALS

Customer:
Part Number:
Panel Size:

ELECTRONIC MATERIALS

Cells with yellow highlight may need data entered.

Number of Layers: 16

Calculated Thickness: 203.06

Prepreg Style	Prepreg Quantity	Prepreg Style	Prepreg Quantity	Design Thickness	Thickness (mils)	Stackup	Layer Number	Layer Type	Copper Weight (ounces)	Estimate of Copper area remaining	Arlon Part Number
							1	Top	0.5	1	
84H 106 80	2	85H 1086 63	2	10	9.08	Prepreg					
							2	Mixed	4	0.54	
				8	8.00	Laminate core					85HP.008 4/4
							3	Mixed	4	0.54	
84H 106 80	4	85H 1086 63	1	10	9.00	Prepreg					
							4	Mixed	4	0.54	
				8	8.00	Laminate core					85HP .008 4/4
							5	Mixed	4	0.54	
84H 106 80	4	85H 1086 63	1	10	9.00	Prepreg					
							6	Mixed	4	0.54	
				8	8.00	Laminate core					85HP .008 4/4
							7	Mixed	4	0.54	
84N 106 80	4	85H 1086 63	1	10	9.32	Prepreg					
							8	Mixed	4	0.54	
				8	8.00	Laminate core					85HP .008 4/4
							9	Mixed	4	0.54	
84H 106 80	4	85H 1086 63	1	10	9.00	Prepreg					
							10	Mixed	4	0.54	
				8	8.00	Laminate core					85HP .008 4/4
							11	Mixed	4	0.54	
84H 106 80	4	85H 1086 63	1	10	9.00	Prepreg					
							12	Mixed	4	0.54	
				8	8.00	Laminate core					85HP .008 4/4
							13	Mixed	4	0.54	
84N 106 80	4	85H 1086 63	1	10	9.32	Prepreg					
							14	Mixed	4	0.54	
				8	8.00	Laminate core					85HP .008 4/4
							15	Mixed	4	0.54	
84N 106 80	2	85H 1086 63	2	10	9.24	Prepreg					
							16	Bottom	0.5	1	

Arlon's In-House Testing Capabilities

Arlon's In-House Analytical capabilities include:

- TMA and DSC glass transition temperature cure checks
- Cross section analysis of laminates and PWB's
- Resin Rheology
- Prepreg flow tests
- FTIR
- Electrical, thermal and mechanical characterizations of resin, glass and copper
- Research and development of new products for future laminate requirements







Property	Method	Value	Typical 85HP Value
Tg (TMA)	IPC 2.4.24C	TMA	>250°C
T300	IPC 2.4.24C	Minutes	>60
CTE X/Y	IPC 2.4.41	ppm°C	17/17
CTE <Tg	IPC 2.4.24C	ppm°C	38.8
CTE >Tg	IPC 2.4.24C	ppm°C	154
Z-axis Expansion (50-260°C)	IPC 2.4.24C	%	≤1

ARLON ELECTRONIC MATERIALS

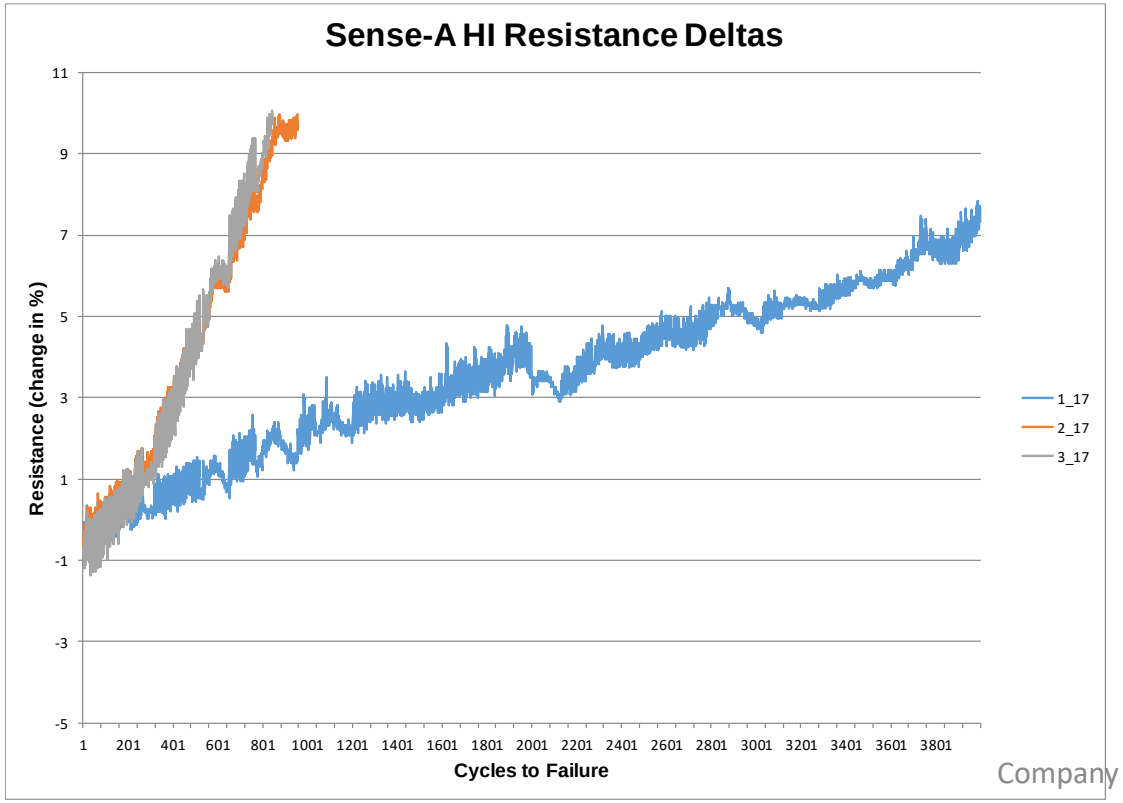
Interconnect Stress Testing

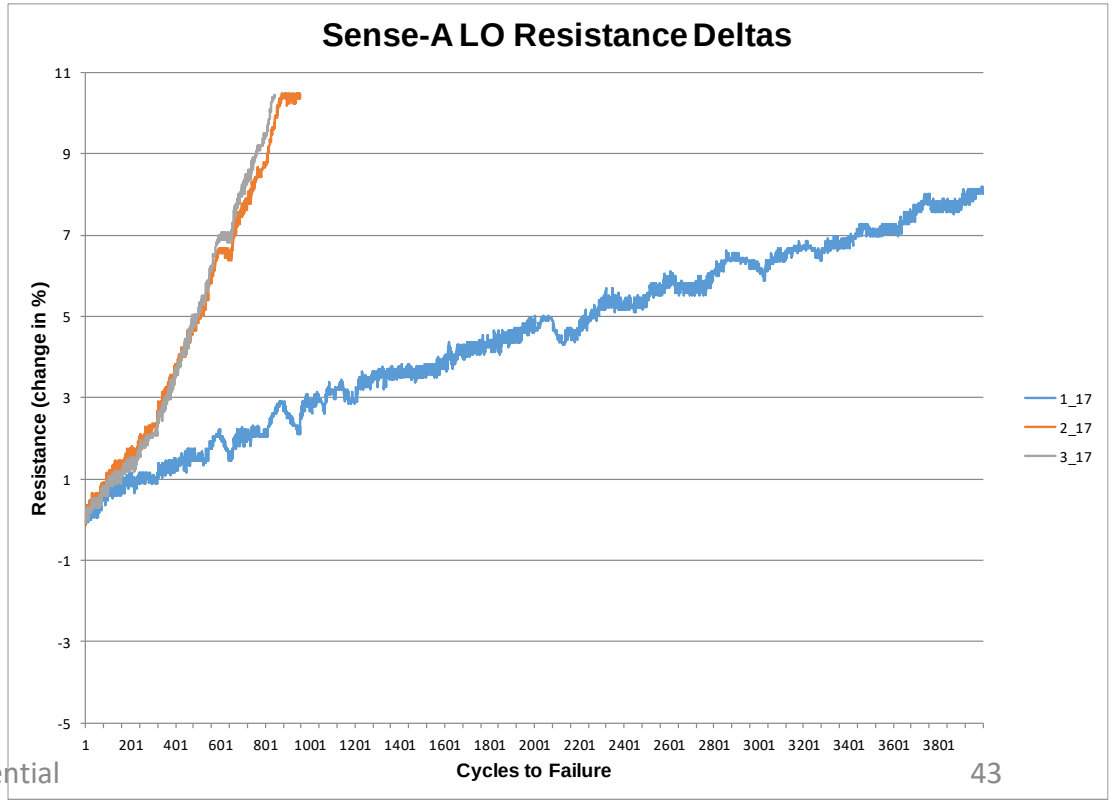
Company Confidential

DC Current Induced Thermal Cycling Test

IST Results: (Compared to 500 cycles on lower Tg products), **85HP was cycled to 4000 cycles 50C - 210°C** (IST programed to 4000 cycles max)

- Blue curve all 85HP. No failure after 4000 cycles!
- Gray and orange curves are a combination of subs of 85N and 85HP. ≈900 cycles (Coupons were run with both 85N and 85HP to compare drilled via holes of each product.





85HP HATS Testing Results

(Highly Accelerated Thermal Shock)

HATS- Highly Accelerated Thermal Shock

- Arlon 85HP 92 mil thick MLB tested up to 1000 cycles from + 145°C to - 40°C
- Total cycle time between temperatures cycling was 10.02 minutes
- All samples were preconditioned with six times reflow cycles up to 260°C
- Twelve coupons tested; all coupons passed thermal shock
- Coupon Design:
  - 18 layer 85HP PWB
  - 2 mil soldermask both sides
  - 8 mil minimum clearance
  - Plane layers; 2,3,6,8,9,10,11,13,16,17
  - Ground plane relief- 18 mils
  - 14 mil hole sizes
  - 33 mil land sizes
  - Track width- 5 mils
  - Grid size- 60 mils

Summary of Arlon's 85HP

- Lowest Z-axis expanding polyimide resin system on the market, (<1% between 50°C-260°C).
- Meets IPC-4101E slash 40, 41 and **43.**
- Pure polyimide, no secondary resin system blended or reacted
- Time to delamination is greater than 60 minutes at 300°C (T300)
- Improved drilling without hole wall resin fracturing typically found with standard PI resins
- Increased drill hit count (8 mil drill yielded 700 hits with 6 tenths of wicking)
- Finished boards meet or exceed the requirements of IPC-6012D maximum of 80 microns (3 mils), IPC 6012DS has a maximum of 50 microns (2 mils) of wicking allowed
- Eliminates neat resin areas adjacent to copper lands our clearance areas (CTE of neat resin is 40 ppm)
- Designed to meet the coefficient of thermal expansion of copper
- Highest thermal conductivity of any other polyimide laminate system on the market
- High resistance of wicking and resin cracking, even and controllable etchback during processing
- Capable of several sequential lamination cycles
- Consistent thickness control, uniform registration using balanced glass styles
- Predictable electrical stability across the plane due to balanced glass styles

Thank you!
&
Questions

www.arlonemd.com